

University of Dhaka
Affiliated Engineering Colleges
B. Sc. in Computer Science & Engineering
2nd Year 1st Semester Final Examination-2021
CSE 2103 (Digital Electronics & Pulse Technique)

Time: 3.00 Hours

Marks: 70

[Answer any Five questions]

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| 1. a) | What is Switch? Draw a two-input NAND and NOR gate circuit using diode. | 6 |
| b) | Define Transistor. "Diode can be used as a switch"- show graphically with characteristic curve. | 4 |
| c) | Explain RTL basic circuits. | 4 |
| 2. a) | Explain classification of logic family. | 3 |
| b) | Draw the Transistor Transistor Logic (TTL) circuit and explain working procedure. | 5 |
| c) | What is pulse transformer? Write the functions of pulse transformer. | 6 |
| 3. a) | What do mean by LED and LCD? | 4 |
| b) | Explain Sample and Hold (S/H) circuit by using Operational Amplifier. | 5 |
| c) | Describe Analog to digital Converter circuit. | 5 |
| 4. a) | Write short Note on- i) Propagation delay ii) Noise margin | 4 |
| b) | Define filtering. Why filtering needed in linear wave shaping? | 4 |
| c) | Explain high pass filter circuit with proper diagrams. | 4 |
| d) | Sketch the frequency response of low pass filter. | 2 |
| 5. a) | Define clipper and clamper circuits. Write down the applications of clipping circuit. | 5 |
| b) | Define free running multivibrator. Describe the working principles of positive clipping circuit with proper diagrams. | 5 |
| c) | Differentiate between negative diode clipping and negative biased diode clipping circuits with figures. | 4 |
| 6. a) | Draw the block diagram of a D/A converter and explain its operation. | 6 |
| b) | Mention the process of triggering of a bistable multivibrator with proper diagram. | 4 |
| c) | Define PLA's working principles. Write down the applications of S/H circuits. | 4 |
| 7. a) | What is Cache Memory? Differentiate between primary and secondary memory. | 4 |
| b) | Explain pulse generator block diagram with equivalent circuit. | 5 |
| c) | What is OP-AMP. Why OP-AMP is a differential, voltage amplifier with high gain? | 5 |